Kyle R. Robin

Stowe, VT | krobin@bates.edu | 917-741-8845 | linkedin.com/in/kyle-robin-931415425 | krobin2005.github.io/myportfolio | U.S. Citizen

Education

Bates College — Lewiston, ME
B.S., Major: Digital & Computational Studies (declared); Intended Second Major: Psychology. GPA: 3.71 — *Expected May 2029*
Relevant Coursework: Intro CS: Software Development (Python), Data Structures & Algorithms*, Evolutionary Computing*, Principles of Psychology* *(*in progress)*

Northeastern University, The Roux Institute (Graduate School) — Portland, ME
Graduate course, taken as an undergraduate through Bates: CY 5010 Cybersecurity Principles and Practices — *Spring 2027*

Burke Mountain Academy — East Burke, VT
High School Diploma. GPA: 4.00 — *May 2024*

Experience

Lead of AI and Machine Learning Research — September 2026 – Present
Castro Lab (Neuroscience), Bates College — *Lewiston, ME*

- Lead AI for Smell under Professor Jason Castro: a six-agent Python research lab that trains PyTorch neural networks to predict odor from molecular structure across 5,858 molecules (Leffingwell + GoodScents).
- Built an evolutionary architecture search whose best network reached 0.837 test AUROC on scaffold splits, outperforming my re-implementation of the graph neural network of Lee et al. (*Science* 2023) on all 5 splits.
- Built a Skeptic agent (scaffold splits, shuffled-label controls, multiple-comparison-corrected bootstrap intervals, a sealed vault set); in a planted-fake test it cut false discoveries accepted from 65% to 25%.
- Built a Claude agent that reads the literature, designs controlled experiments, and passes a checker that rejects fabricated citations. Working toward a co-authored paper.

Technology Sector Analyst — August 2025 – Present
Bates Investment Club — *Lewiston, ME*

- Build investment theses on technology companies (previously industrials) using DCF and comparable-company analysis; pitch stocks to the club.

Greenskeeper — June 2024 – August 2026 (Summers)
West Hill Golf Course — *Stowe, VT*

- Led turf, irrigation, and course-design projects on a four-person grounds crew; operated heavy equipment under safety and environmental rules.

Co-Lead Researcher and Project Analyst — January 2024 – May 2024
BMA Energy Proposal — *East Burke, VT*

- Co-led a $1.6M solar feasibility study covering 248,000+ kWh of annual campus electricity use, evaluating system sizing, project economics, and federal and Vermont incentives (IRA Elective Pay, ITC, PTC).

Projects

Strayline: Founder, AI Network-Attack Detection Venture — October 2026 – Present

- Built a hybrid intrusion detector in PyTorch: a classifier for known attack families plus an autoencoder trained only on benign traffic. A leave-one-family-out test on CIC-IDS2017 maps which never-seen attacks are caught at a 1% false-alarm rate (e.g., 87% of unseen infiltration).
- Added three Claude agents (triage, an investigator using an MCP tool server, and a literature researcher) on pseudonymized, label-free incidents: 49,387 alerts became 414 incidents and 55 escalations, with all 8 real attack incidents escalated, including a never-seen port scan.
- Submitted as a research proposal to the AAAI-27 Undergraduate Consortium; applied to pitch at The Eleven Society's Northeast Entrepreneurship Summit (Boston, Nov. 2026).

Scent to Art: Prize Winner, Bowdoin–Bates–Colby Hackathon — September 2026

- Built a five-agent Claude API pipeline that proposes a structural change to a molecule and predicts its new odor profile; presented to ~100 attendees, including the president of Bowdoin College.

Athletics

NCAA Division I Alpine Skiing, Bates College — August 2025 – Present

- Balance 30+ hours per week of training and competition while maintaining a 3.71 GPA.

Internationally Competitive Ski Racer, BMA E-Team — May 2024 – April 2025

- Sponsored athlete in FIS, National Championship, and Continental Cup races across North America and Europe (Austria, Italy, Slovenia, Canada); earned FIS victories and top-10 National Championship results.

Skills

Machine Learning: Neural networks (MLPs, autoencoders, GNNs), evolutionary search, anomaly detection, AUROC and bootstrap evaluation
Programming: Python (PyTorch, scikit-learn, pandas, NumPy, RDKit, Matplotlib), R, HTML/CSS/JavaScript, Git and GitHub
AI: Claude API and Claude Code (multi-agent systems, tool use, MCP servers), LLM evaluation, ChatGPT, Gemini, Codex
Finance: DCF, comparables, accounting · **Software:** Excel, Word, PowerPoint · **Certifications:** Anthropic AI Fluency